Retell AI Topic 7 - Custom Function Calling

Lesson 5 Assessment: NovaNest create_booking_request

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GitHub repository	https://github.com/shaikshahid777/novanest-custom-booking-function
Demo video (Loom)	https://www.loom.com/share/a227ccdad81945b5abc511efccc75fcd
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1. Project Summary

This is a mock booking-request workflow: the booking reference is simulated and no real calendar appointment is created. The NovaNest Consultation Booking Agent (Retell AI) was extended with one custom function, **create_booking_request**. After the caller provides all booking details and says yes to "Shall I go ahead and submit this booking request now?", the agent sends the request to my own Node.js mock endpoint through an ngrok tunnel. The endpoint validates the data, returns a booking reference, and the agent reads it to the caller. If the endpoint fails, the prompt tells the agent to give a safe fallback message without reading error codes.

2. Function Configuration (Retell)

Setting	Value
Name	create_booking_request
Method / URL	POST - https://unimpeded-shifter-statute.ngrok-free.dev/create_booking_request
Header	ngrok-skip-browser-warning: true
Timeout / Max retries	10 seconds / 2
Payload	Args only: ON (parameters at top level of body)
Talk while waiting	ON - "Let me go ahead and submit that booking request for you now."
Talk after action completed	ON
Parameters (JSON schema)	caller_first_name, caller_last_name, caller_phone (10 digits), caller_email, service_type (Buyer Consultation / Seller Consultation / Investment Review), preferred_window_1 (required), preferred_window_2 and preferred_window_3 (optional)

3. Prompt Additions

- **Path 1, steps 7-9:** ask permission to submit, wait for a clear yes, then call the function.
- **Function call rules:** new bookings only; no call before all required fields and consent; phone as digits only; service_type exactly as listed.
- **Function result handling:** on success read the reference naturally; on error or no response say "I wasn't able to submit the request right now, but your details have been noted and our team will follow up shortly to confirm your slot."; never read error codes, field names or JSON keys.

4. Mock Endpoint (server/server.js)

- Missing required field returns HTTP 400 with MISSING_FIELD; invalid format returns HTTP 400 with INVALID_FORMAT.
- Valid request returns HTTP 200 with a booking reference (format NNB-YYYYMMDD-xxxx).
- Idempotent: the same payload returns the same booking reference and creates no duplicate.
- Test modes: ?mode=503 (forced service error) and ?mode=delay (15-second delay).

5. Test Results

Test	Layer	Scenario	Result	Evidence
TC-001	Retell	Happy path; function fires after "Yes, go ahead"	COVERED	Loom video; server log shows request returning NNB-20261010-6547

Test	Layer	Scenario	Result	Evidence
TC-002	Retell + log	Correct parameters at endpoint	PASS	Server log (screenshots/server-log-happy-path.txt) shows all parameters at top level (Args only); ngrok shows 200 OK
TC-004	Retell	Reference read naturally	COVERED	Loom video
TC-007	Endpoint	Missing phone	PASS	PowerShell endpoint test screenshot: HTTP 400, MISSING_FIELD
TC-008	Endpoint	Invalid phone ("123")	PASS	PowerShell endpoint test screenshot: HTTP 400, INVALID_FORMAT
TC-009	Endpoint	Idempotency	PASS	PowerShell endpoint test screenshot: both requests returned NNB-20261010-1296
TC-003	Retell	Early trigger prevention	NOT RUN	-
TC-005	Retell	503 handling	NOT RUN	-
TC-006	Retell	Timeout and retry handling	NOT RUN	-
TC-010	Retell	No raw error data spoken	NOT RUN	-

Totals: 4 PASS, 2 COVERED (in the demo recording, not individually verified), 4 NOT RUN. Failure and timeout behaviour was verified at the endpoint level only (TC-007, TC-008, TC-009); the Retell-side failure tests were not executed.

6. Assumptions, Limitations and Challenges

- **Assumptions:** Topics 1-6 are in place; the mock endpoint stands in for a production API (switching is a URL change).
- **Limitations:** idempotency store is in memory and resets on restart; no endpoint authentication; no real CRM or calendar integration; booking reference is simulated; ngrok free URL can change.
- **Challenges:** the ngrok free-plan browser warning page (fixed with the ngrok-skip-browser-warning header); keeping the agent from calling the function before consent (explicit trigger rules in the prompt); matching test numbering to the requirement documents.

7. Possible Enhancements

- Endpoint authentication; database-backed idempotency and booking storage.
- Real CRM or calendar API integration; SMS or email confirmation function.
- Run and record the Retell failure tests (early trigger, 503, timeout); add automated tests.

These enhancements are not implemented in this submission.